

From Concepts to Measurement: Understanding Variables in Social Science

POLI 110 Tutorial 04

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1 The Framework: Concepts, Variables, and Measures

1.1 Key Definitions

- **Concept:** An abstraction used to describe characteristics of a phenomenon
 - Cannot be proper nouns
 - Examples: democracy, state capacity, political participation
- **Variable:** A measurable property that can vary across cases or time
 - Maps onto concept of interest
 - Must be observable
 - Examples: electoral turnout, GDP per capita
- **Measure:** A procedure for determining variable values
 - Must be systematic
 - Must be replicable
 - Examples: survey questions, counting procedures

2 Types of Variables and Measurement

2.1 Nominal/Categorical Variables

- Categories with no natural order
- Example concepts:
 - Political system type
 - Electoral system
 - State religion
- Key feature: Categories are mutually exclusive

2.2 Ordinal Variables

- Categories can be ordered but distances unknown
- Example concepts:
 - Level of democracy
 - Strength of civil society
 - Degree of press freedom
- Key feature: Meaningful ranking possible

2.3 Interval/Ratio Variables

- Precise numerical values
- Example concepts:
 - Population size
 - Economic indicators
 - Election vote shares
- Key feature: Meaningful distances between values

3 The V-Dem Dataset

3.1 Overview

- Website: <https://v-dem.net>
- Data access: <https://v-dem.net/data/the-v-dem-dataset/>
- Online analysis: <https://v-dem.net/analysis/>
- Codebook: <https://v-dem.net/data/reference-documents/>

3.2 Dataset Features

- Time period: 1789-2022
- Countries covered: 202
- Variables: Over 500 indicators
- Types of measures:
 - Expert coded variables
 - Composite indices
 - Factual indicators

4 Exercise: Variable Analysis in V-Dem

4.1 Task: Finding Different Types of Variables

For each type of variable (nominal, ordinal, and interval/ratio):

1. Find one example in V-Dem
2. Identify what concept it measures
3. Explain why it uses that level of measurement

4.2 For Each Variable Consider

- **The Concept**
 - What is being measured?
 - Why is this concept important?
 - Is it a direct or indirect measure?
- **The Measurement**
 - Why this level of measurement?
 - What are the limitations?
 - Could it be measured differently?

4.3 Discussion Questions

1. How do different types of variables capture different aspects of democracy?
2. What are the advantages and disadvantages of each measurement level?
3. How does the choice of measurement level affect our analysis possibilities?

5 Example Analysis

Variable Example: Electoral Democracy Index (v2x_polyarchy)

- **Type:** Interval (0-1 scale)
- **Concept:** Quality of electoral democracy
- **Why this level?**
 - Allows fine-grained comparisons
 - Enables statistical analysis
 - Captures subtle changes over time